

- (c) Electromagnetic (d) Polarized
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Sound waves in air (and in any fluid medium) are longitudinal waves because particles of the medium through which sound is transported, vibrate parallel to the direction that the sound wave moves.

3. Cosmic rays –

- (a) Are charged particles
(b) Are uncharged particles
(c) Can be charged as well as uncharged
(d) None of the above

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Cosmic rays are not the part of electromagnetic spectrum. They are immensely high-energy charged particles, travelling through space at a speed close to speed that of light. They originated either from the sun or outside of our solar system.

4. What is the distance between two successive crests or successive troughs called?

- (a) Amplitude (b) Wavelength
(c) Frequency (d) None of these

45th B.P.S.C. (Pre) 2002

Ans. (b)

Wavelength means the distance measured in the direction of a wave from any given point to the next point in the same phase, as from crest to crest. The distance between two successive crests or two successive troughs is the wavelength of a transverse wave.

5. The velocity of electromagnetic waves is :

- (a) $3 \times 10^8 \text{ ms}^{-1}$
(b) $3 \times 10^7 \text{ ms}^{-1}$
(c) $3 \times 10^6 \text{ ms}^{-1}$
(d) $3 \times 10^5 \text{ ms}^{-1}$
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The velocity of all electromagnetic waves is the same and all travel at the equal speed of light in a vacuum. Hence, the velocity of electromagnetic waves is about $3 \times 10^8 \text{ ms}^{-1}$.

6. Which of the following is not electromagnetic?

- (a) Gamma rays (b) Alpha rays

- (c) X-rays (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (b)

Electromagnetic waves include visible light waves, X-rays, gamma rays, radio waves, microwaves, ultraviolet and infrared waves. The electromagnetic spectrum includes both non-ionizing and ionizing radiation. All types of electromagnetic radiation, ionizing and non-ionizing, travel at the speed of light. Their energies are not determined by their speed (the speed of light) but by their frequencies (the number of waves or cycles per second). Alpha and beta particles are not part of the electromagnetic spectrum. They travel slower than the speed of light.

7. Which of following rays are used in the diagnosis of intestinal diseases :

- (a) X-rays (b) α -rays
(c) β -rays (d) γ -rays

43rd B.P.S.C. (Pre) 1999

Ans. (a)

X-rays are a form of electromagnetic radiations, which are used in the diagnosis of intestinal diseases.

8. The Sonar technique is used

- (a) to locate icebergs
(b) to locate underwater hills
(c) to determine the depth of the sea
(d) All of the above

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

Sonar stands for Sound, Navigation and Ranging. Sonar technique is used for exploring and mapping the ocean because sound waves travel farther in water than do radar and light waves. It can detect the echo of any sound and detect it. It is used to detect icebergs, to locate underwater hills, to determine the depth of the sea.

Sound

1. A noise level of 100 decibel would correspond to

- (a) just audible sound
(b) ordinary conversation
(c) sound from a noisy street
(d) noise from a machine shop
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)